

The Synthesizer

Isaiah Mak | March 10, 2026 | Last modified April 13, 2026

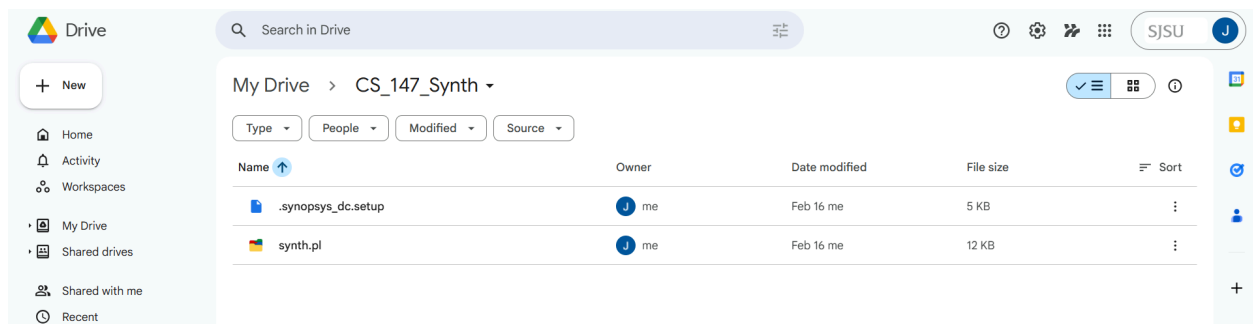
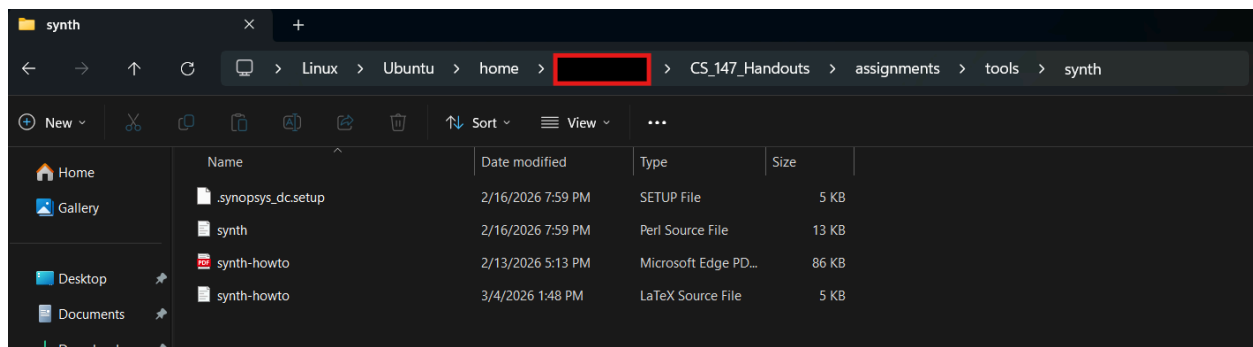
Credit: Original instructions written by Nainoa Jackson and Dr. Sriram Rao. Windows screenshots credited to Jayden Dieu. Screenshots for Mac, Ubuntu, or other can be sent to Isaiah to include.

MAKE SURE YOU FOLLOW ALL STEPS EXACTLY OR SOMETHING MAY NOT WORK.

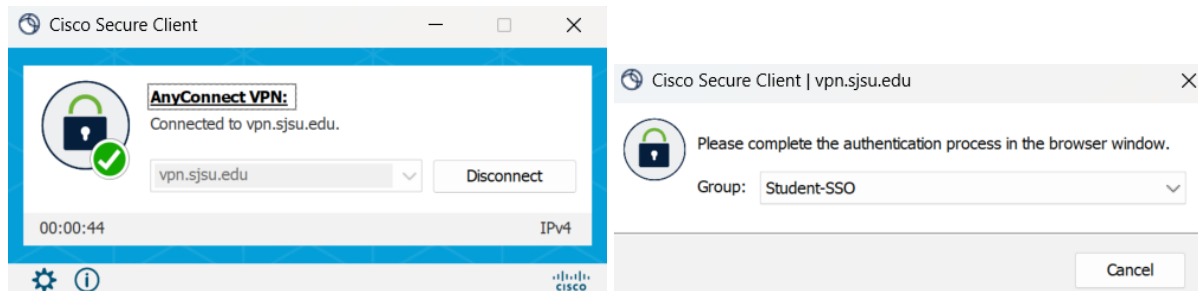
Setting up your CAD machine

*From here, things are done on your **local computer**.*

1. Create a folder in a cloud service so you can be organized. I made my folder called "Lab (to copy to CAD)"
2. Find the repo clone (CS_147_Handouts) in File Explorer or Finder (or whatever).
3. Upload **assignments/tools/synth/.synopsys_dc.setup** to a cloud service such as Google Drive.
4. Upload **assignments/tools/synth/synth.pl** to a cloud service such as Google Drive.



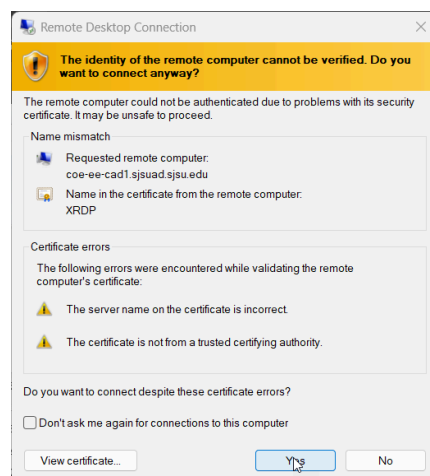
5. Connect to the VPN via Cisco Secure Client. The SJSU VPN is **vpn.sjsu.edu**. Make sure you are logging in as a Student-SSO (if you're a student obviously).



6. Open the remote desktop feature to login to the CAD machines. For Windows, this is Remote Desktop Connection. For MacOS, you need to download the Microsoft “Windows App”. For Ubuntu, you need to use a service called Remmina → “sudo apt update && sudo apt install remmina remmina-plugin-rdp”

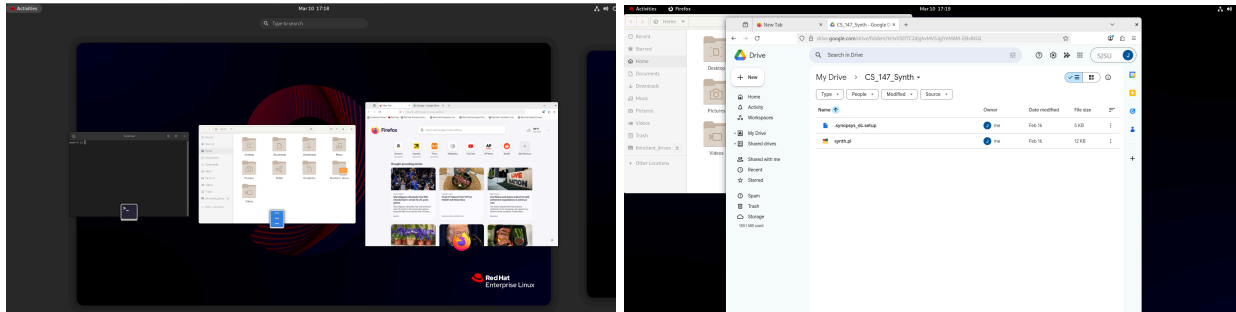


7. When it asks for the computer, type in **coe-ee-cadXX.sjsuad.sjsu.edu** where XX is a number between 1 and 50 (omit the 0 if single digit). It will prompt that the remote computer cannot be verified (at least on Windows for me). Select to connect anyway by clicking **Yes**.

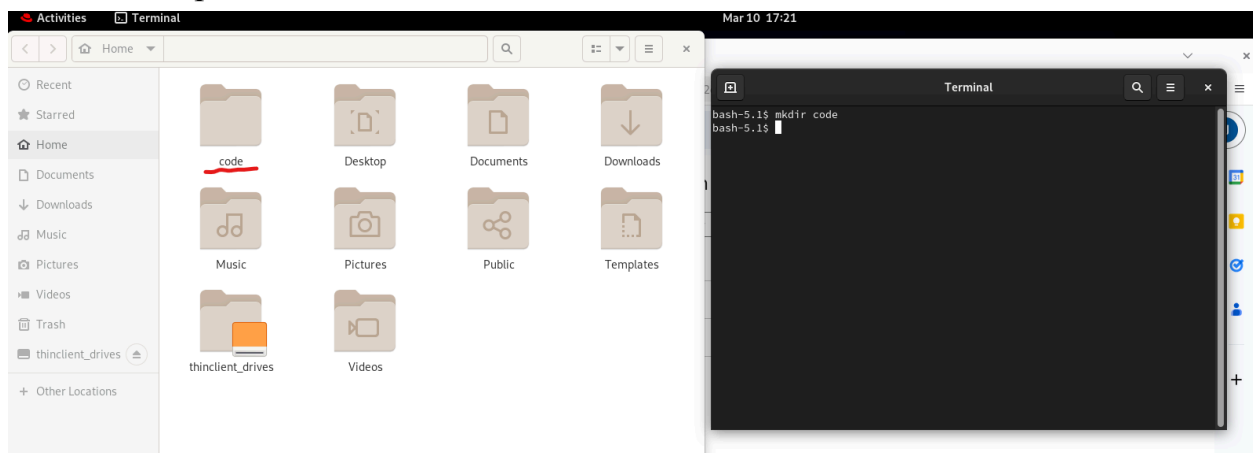


*From here, things are done inside the **remote computer**.*

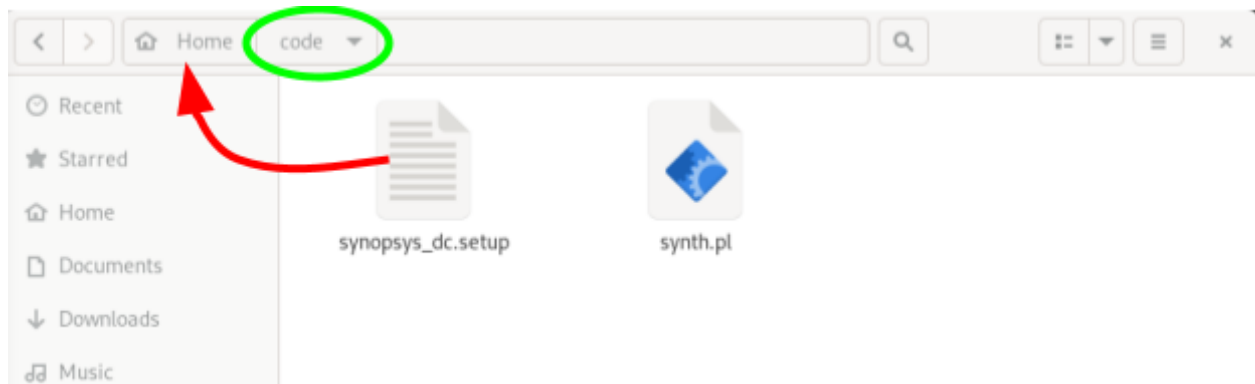
8. It will bring you to an SJSU login. It is the same as your SJSU Okta (SJSU ID and password). If it's your first time or it has been a while since your last login, there may be a black screen for some time. After that, it might also bring you to another login. For that, do the same thing: use your SJSU password.
9. Open Firefox and Files using the Activities button on the top left. On Firefox, go to your cloud service.



10. Make a new folder, which will hold your imported labs, in the Home directory of the remote computer. I used `code` since that's what the original writeup used.

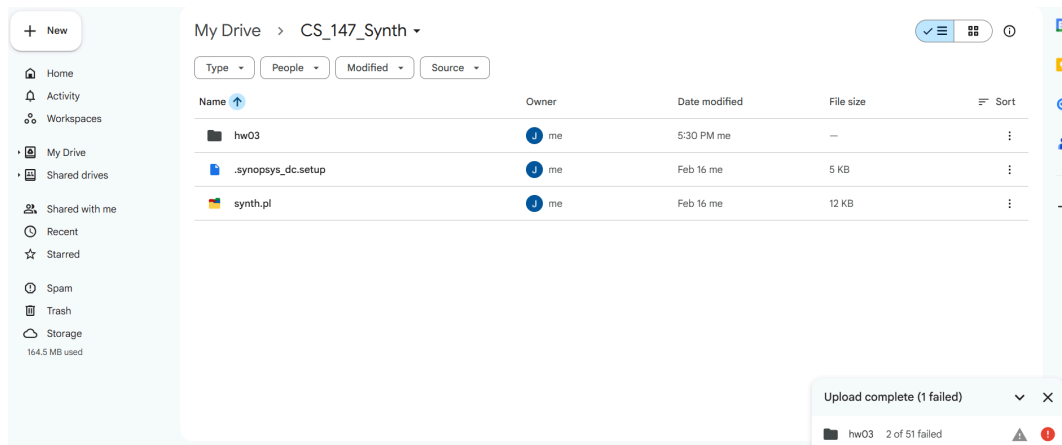


11. Download **.synopsis_dc.setup** to the Home directory. In cases I have seen, when you download it, the beginning "." will be removed. For example, **/home/.synopsis_dc.setup** is where the file should be.
12. Rename **synopsis_dc.setup** with a "." at the beginning so that it is **.synopsis_dc.setup**. Also, this will make the file hidden.
13. Download **synth.p1** to the folder you just created (`code` in my case).

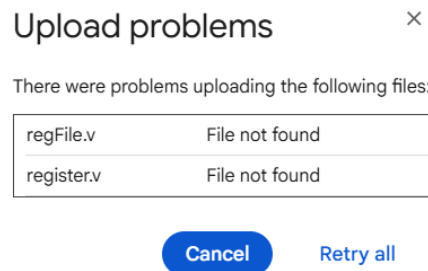


How to use the Synthesizer

1. In your local computer, upload the lab that you want to test to your cloud service. Upload the *entire* lab for convenience if you need its multiple parts. For example, upload hw03 instead of just hw3_2.



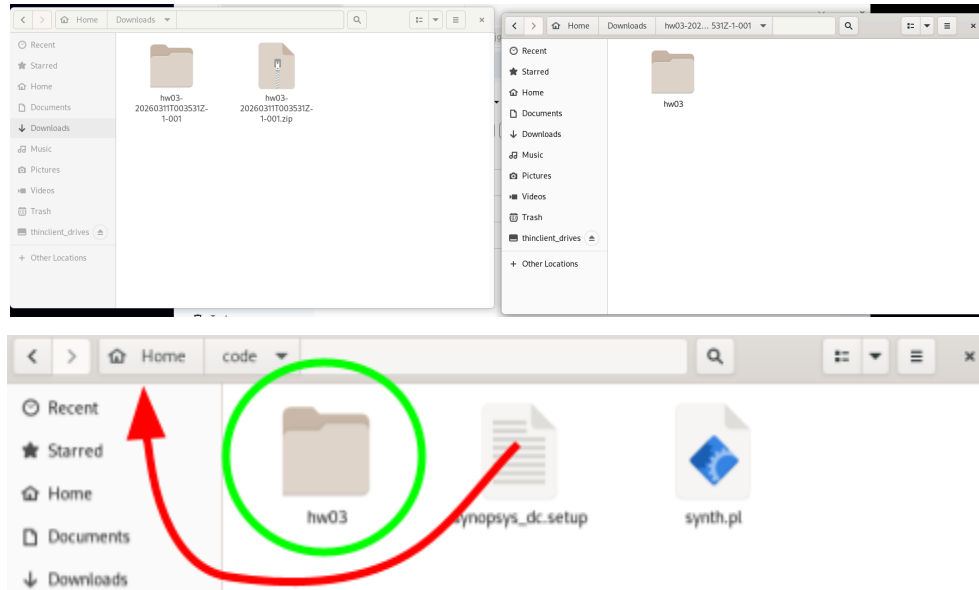
2. Important Note for Lab 3: Since register.v and regFile.v are connected by symlink, it will fail to download those files in hw3_2. Fear not. Just copy and paste those files from hw3_1.



From here, things are done inside the **remote computer**.

3. Download your lab. It will have an extended name and will be a zip. Double-click it to get an unzipped version. Inside the unzipped version, you

will see the folder that has your project. Move it (`hw03`) into the folder you created (again, `code` in my case).

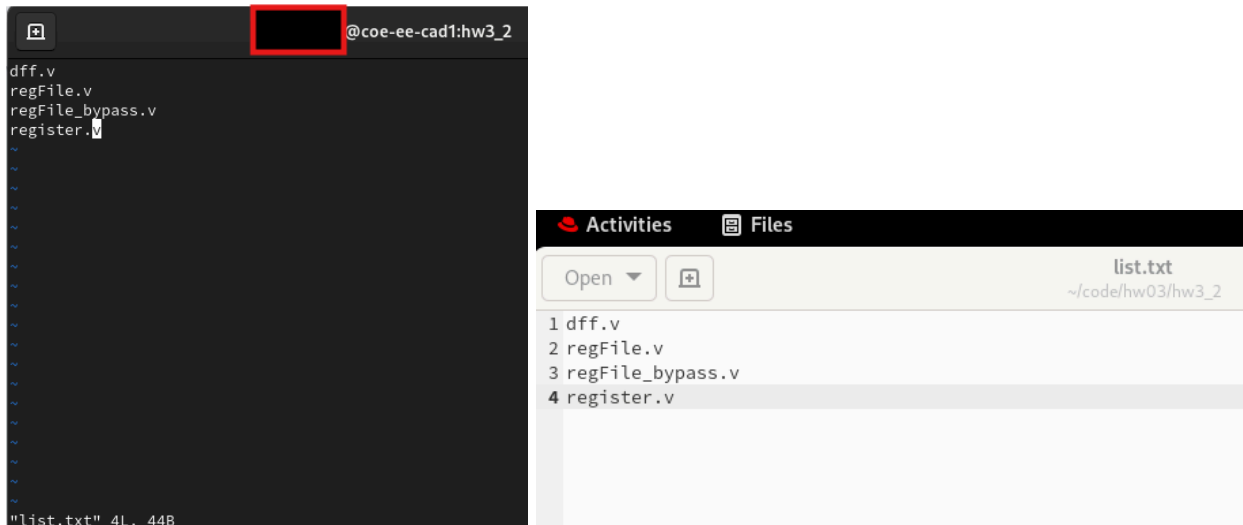


Note: From here, note that a lot matches with part 3 of `synth-howto.pdf`.

4. (Step 3.3) Create a text file named `list.txt` inside the exact problem (and directory) that you are running. This means exactly:

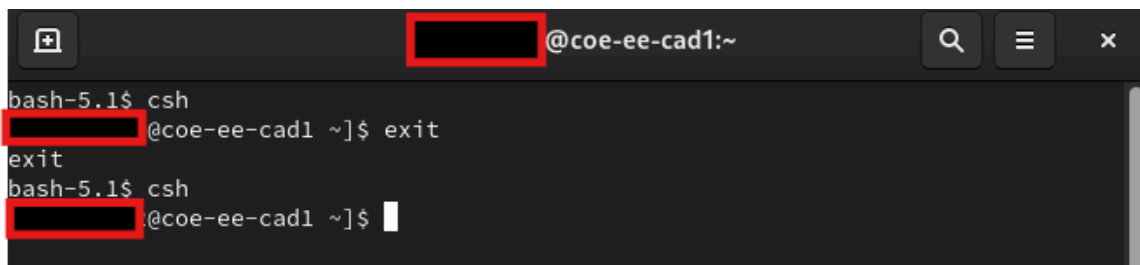
`/code/hw03/hw3_2` for the Lab 3 example. You can do this using `vi` or go to Activities on the top left and find Text Editor (in the waffle). Fill this document with the names of all the *Verilog* files that ***you used***, one per line.

- a. **No** testbenches should be included in this list
- b. **No** `_hier` files which are wrappers should be in this list
- c. `clkrst.v` should **NOT** be in the list, since we don't want to synthesize the clock generator



5. (Step 3.2) Open the command prompt, type these two commands:

```
> csh
> source /apps/design_environment.csh
```



6. (Step 3.4) On the command prompt and head to the directory *that you made your list.txt file* (using `cd`). Check the design by running the command below. In other words, check if it works.

- Important: `list.txt` is just the name of your text file holding the Verilog files. You can technically name it whatever you want.
- Important: `--top=regFile_bypass` uses `regFile_bypass` because it is the “most top-level” Verilog file. It will work with other names, but it may not give you all the information you need. For example, the project would likely want `proc` for this.

```
> cd code/hw03/hw3_2
> perl ~/code/synth.pl --list=list.txt --type=other
--cmd=check --top=regFile_bypass
```

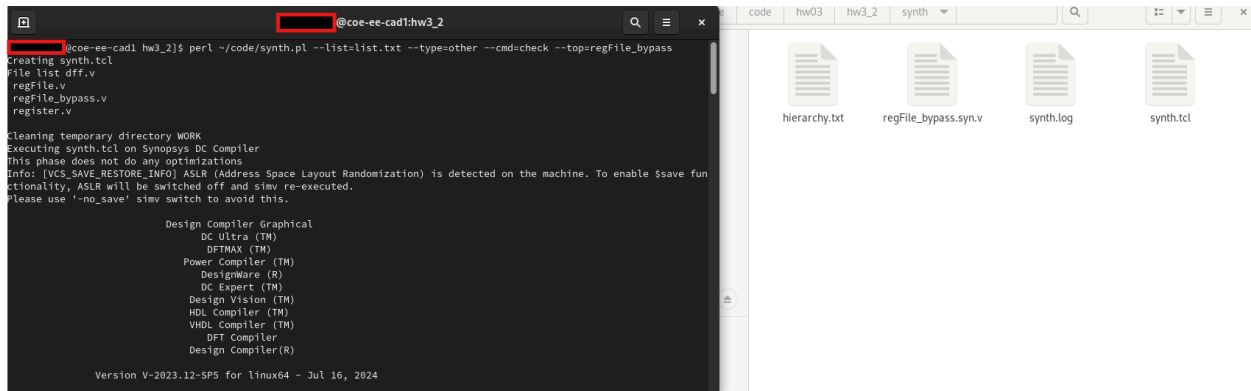
7. Make sure that the output, which is also going to be in `synth.log`, has **ABSOLUTELY** no errors. Do not proceed unless that is true. Most Warnings can be ignored, but please refer to this:

<https://pages.cs.wisc.edu/~sinclair/courses/cs552/spring2023/synthesis-faq.html#toc7>

8. (Step 3.5) Run the synthesis. (The only change was on `--cmd=`)

```
> perl ~/code/synth.pl --list=list.txt --type=other
--cmd=synth --top=regFile_bypass
```

9. There should be a folder named `synth` in the directory that you ran the command in. Inside, there are files named `area_report.txt`, `cell_report.txt`, and `timing_report.txt`. These files have some things you need to check before your submission. For example, part 3 of Lab 3 tells you to check two of these files.



10. If you want to optimize the design, add a param to the end `--opt`

```
> perl ~/code/synth.pl --list=list.txt --type=other
--cmd=synth --top=regFile_bypass --opt
```

11. For Lab 3 (and likely for subsequent labs), you will need to upload the `synth` folder to your cloud service. In your local computer, download the `synth` folder and copy it to the respective folder. Again, specifically **/assignments/hw03/hw3_2** for the Lab 3 example. Depending on your experience with Linux commands, your methods may vary.

- a. If you know how to use `mv`, you can try that. However, from my experience, I don't have write permissions to the repo clone. This means the `mv` may fail.
- b. Add the **synth** folder into `/hw03_submission/hw3_2`. Then run the submission command: `./run build_submission <project>`. Again, from my experience, this may fail as well.
- c. My best fix for the submission is the following:
 - i. Do the submission command first: `./run build_submission <project>`
 - ii. Copy the submission zip folder out of the repo clone. I just moved it to the Downloads folder because yes.
 - iii. Copy or move the **synth** folder into `/hw03_submission/hw3_2` *without unzipping*.